



Parkinson's disease

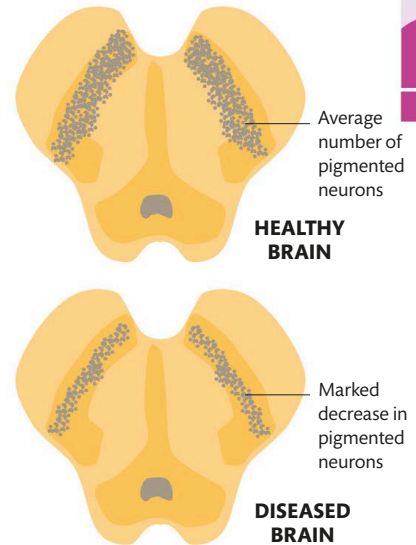
The second-most common degenerative disease after Alzheimer's (see p.50), Parkinson's disease is a neurological disorder that affects movement and mobility by destroying dopamine-producing cells in the substantia nigra, which is located in the uppermost part of the brain stem.

CAN SURGERY BE USED TO TREAT PARKINSON'S DISEASE?

Deep-brain stimulation (DBS) involves surgical implantation of electrodes in the brain that can control, but not cure, the motor symptoms of Parkinson's.

Symptoms manifest gradually, sometimes starting as a mild tremor in one hand. Other signs include muscle stiffness, slurred speech, and a general slowing of mobility. Early stages of the disease usually affect one side of the body, but when 80 percent of the substantia nigra dies, severe disablement occurs. Late-stage sufferers require assistance with all daily tasks.

Parkinson's mainly strikes adults aged 60 or over and affects more men than women.



Changes in the substantia nigra

Parkinson's affects nerve cells in the substantia nigra, which produce the neurotransmitter dopamine. As the cells die, dopamine levels fall, disrupting motor control.

Huntington's disease

Huntington's is a progressive brain disorder caused by a genetic mutation. Early signs include irritability, depression, involuntary movements, poor coordination, and trouble with decision-making or learning new information.

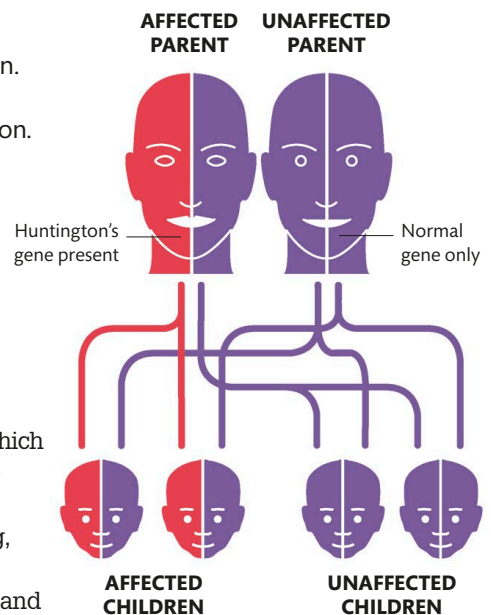
Adult-onset Huntington's is the most common form of the disease, and it usually appears in people in their 30s and 40s. It affects three to seven out of 100,000 people of European origin. Less frequently, it begins in childhood or adolescence, where it causes mobility problems and mental and emotional changes.

Additional symptoms of juvenile Huntington's include slow movements, clumsiness, frequent falling, rigidity, slurred speech, and drooling. Thinking and reasoning abilities are impaired, which affects performance in school. Seizures occur in 30 to 50 percent of children

with this condition. Juvenile Huntington's disease tends to progress rapidly.

Huntington's chorea

Many people with Huntington's develop involuntary twitching movements known as chorea, which become more pronounced as the disease progresses. They may have difficulty walking, speaking, and swallowing, and may also experience personality changes and a decline in thought processing. Prognosis for people with adult-onset Huntington's is a life span of 15 to 20 years after symptoms begin.



Patterns of inheritance

Huntington's is classed as an inherited condition. It occurs when a single defective gene is passed on from an affected parent.